Chips

"OUR APPROACH INVOLVES Not Only Designing The System But Also CREATING TEST SETUPS For Various Scenarios"

Discover the secrets behind successful IP development as we delve into innovative design strategies. Learn how to navigate the intricate balance between power efficiency and high performance. Sharad Bhowmick and Nidhi Agarwal from Electronics For You spoke to Dr Arun Ashok and Rijin John, Co-founders, Silizium Circuits and got some very interesting details...

(L to R) Rijin John and Dr Arun Ashok, Co-founders

How does Silizium Circuits contribute to IP development?

Yes, we create something called 'hard IPs.' These are like small, important parts inside electronic devices, kind of like tiny chips within a bigger chip. Imagine opening up your electronic device and seeing many tiny chips working together to make it function. Companies like Intel, Renesas, or Infineon want to make their own chips, but stringent time to market necessitates more cooperation and collaboration. So, they seek partners like us who can provide these important parts, or IPs, to help them build their chips faster and better. This saves them time and money.

We are an Indian company, producing IPs that other companies can use in their products. We give them a graphic data system (GDS) file with all the necessary design details. It is a confidential file exclusively for their use, safeguarding our IPs. There are two types of IPs: one resembling the tiny electronic parts mentioned earlier, and the other is a special code called RTL logic. We offer both to assist companies in expediting their product development more efficiently.

What tools and methods do you use to create hard IPs?

In our chip and IP design company, the technologies we use are connected to the foundry nodes that we collaborate with. Foundry nodes determine the smallest dimensions of structures within a chip. Our primary focus lies in analogue and RF technologies, ranging from 65 nanometres and upwards. We can work with nodes like 65nm, 130nm, and 180nm. We have agreements with various foundries enabling us to produce items using these nodes. However, when creating IP for clients, it must align with the technology they use. While we engage with nodes from 65nm and above for our in-house projects, we can also develop IP for clients using smaller nodes like 22nm or 28nm, based on their requirements.